32-bit Pipelined RISC-V Processor Project Summary

Overview

This project implements a learning-oriented 32-bit pipelined RISC-V processor in Verilog HDL. It includes modular RTL blocks, testbenches, simulation using Icarus Verilog, and waveform debugging using GTKWave.

Implemented Modules

• Program Counter
• Instruction Memory
• Control Unit
• ALU and ALU Control
• Register File
• Immediate Generator
• Instruction Fetch Stage
• ID Stage
• Execute Stage
• Memory Stage
• Write Back Stage
• Hazard Detection Unit
• Forwarding Unit
• Pipeline Registers

Tools Used

Ubuntu WSL, VS Code, Verilog HDL, Icarus Verilog, GTKWave, Git, GitHub.

Folder Structure

rtl/, tb/, sim/, docs/.

Compilation

iverilog -g2012 rtl/control_unit.v rtl/riscv_pipeline/*.v tb/riscv_pipeline/pipeline_top_tb.v -o sim/pipeline_top

Simulation

vvp sim/pipeline_top
gtkwave riscv_pipeline_top.vcd

Current Status

Individual RTL modules are implemented and largely verified. Top-level pipeline integration is in progress.

Learning Outcomes

RTL coding, modular design, pipelined CPU architecture, simulation, waveform debugging, Git workflow, and RTL integration/debugging.